

# Does the Assessment Require the Skill?

## Matching released mathematics keys without the tagged operation

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### Abstract

Released mathematics items carry a published tag: a curriculum standard, or a cognitive domain such as reasoning. A correct answer is then read as evidence of that competency in accountability, curriculum evaluation, and research. If a pass does not require the tagged operation, those uses are reading the wrong quantity. We ask whether a program that lacks that operation can still match the published key. A match is one-sided: that pass does not require the tagged operation, and it does not say students failed to learn. Failure to find a program does not certify the tag. Students are not measured.

On public-release items we score three families, each Holm-corrected (a family-wise multiple-testing adjustment) on its own: catalog item-writing rules, adaptive strategies, and partial-input language models. A catalog of item-writing rules does not survive that correction. Two catalog tests beat chance and remain candidates for replication, not family-wise discoveries. Image-channel backsolving (transcribe the printed equation, then substitute options) fires on a small unique-equation subset of Algebra I. A language model recovers Algebra I and Algebra II keys with the given quantities withheld; on those items a pass may require recognizing the form the wording names rather than executing the tagged computation on the givens. The same search finds no catalog, strategy, or partial-input program on Trends in International Mathematics and Science Study (TIMSS) knowing, applying, or reasoning items.

**Keywords:** assessment validity, test-wiseness, TIMSS

### 1 Introduction

An assessment item arrives with a claim about what answering it requires. Utah Core and the Common Core State Standards (CCSS) print a standard code. The Trends in International Mathematics and Science Study (TIMSS) prints knowing, applying, or reasoning [International Association for the Evaluation of Educational Achievement 2009]. A passing score is then used as evidence of that competency: in accountability, in curriculum evaluation, and in research that treats tagged items as operationalizations of skills. If a pass does not require the tagged operation, those uses are reading the wrong quantity. The testing standards treat that score use as a claim that must be validated for the intended interpretation [American Educational Research Association et al. 2014].

That inference is an argument, not a definition [Kane 1992, 2013; Messick 1995]. It can be wrong even when the publisher's tag is sincere. The item may be answerable from a surface cue, from the choices without the stem, or from a formula that never binds symbols to the situation the standard names.

This paper searches for those rival explanations as programs. A program that lacks an operation the tag names, never sees the published key, and still matches the key under the published scoring rule is a *witness*. A witness refutes the claim that a pass on that item requires the tagged operation. A search that finds no witness does not certify the tag beyond the programs that were tried. We do not estimate whether students learned the skill, whether humans use the shortcuts, or which curriculum is better. We never say students do not learn  $X$ .

Results are cell-level pass rates for named programs, not an item-by-operation table. Catalog rules lack content operations. Masked-stem withholds the given quantities. TIMSS items keep knowing, applying, and reasoning. We do not assign CCSS codes to TIMSS items.

Our contributions are the object, the bar, the census, and what the census found.

- **A one-sided test of a tag that needs no student responses.** We turn the claim that an item requires an operation into a falsifiable statement about programs: a program that lacks the tagged operation, never sees the published key, and matches the key under the published scoring rule is a witness that a pass does not require that operation. The test uses no student data and never yields a claim that students failed to learn. Validity theory states the argument; cognitive diagnosis and test-wiseness research estimate students or list cues. Neither scores this object (Section 2).
- **A pre-registered witness bar and family discipline.** A witness must fire on at least 10 items with a lower 95% confidence bound above format chance, adjusted for option elimination; searched and fitted programs must also beat the 95th percentile of random programs of matched complexity. Programs are hashed before scoring. Catalog rules, adaptive strategies, and partial-input language models are three families, each Holm-corrected on its own, so a row that beats chance in one cell is never billed as a family-wise discovery. Before any null is read, the same searcher must recover published shallow-cue rates on public benchmarks (Section 3).

- **A census of tagged public-release mathematics.**

We assemble 2,405 keyed items from nine authorities, each tag quoted from its authority, including the last TIMSS cycle whose items, keys, and domain tags are public together. Channels are defined by the information and operations a program lacks: surface cues, adaptive test-taking strategies, substitution of the options into the printed equation read from the page image, and language models that see the question with every given quantity withheld.

- **What the census found.** The item-writing catalog yields no family-wise discovery; two rows beat chance in their cell and await replication. The one strategy that survived correction failed a powered replication. A language model in two model families recovers New York Regents Algebra I and Algebra II keys with the given quantities withheld, so on those items a pass does not require executing the tagged computation on the givens. Image-channel back-solving fires on a small unique-equation subset of Algebra I. TIMSS knowing, applying, and reasoning stay null at the public  $n$ , with the same searcher validated on benchmarks with documented cues (Section 4).

This census uses public-release items. TIMSS 2011 is the last cycle whose item texts, keys, scoring guides, domain tags, and percent correct are public together, with no permission process. From 2015, the International Association for the Evaluation of Educational Achievement (IEA) keeps most items secured; restricted-use grants typically cap example items and forbid the comparisons this census needs. Recency of showable items comes from released state and provincial tests, not from a later TIMSS cycle.

## 2 Related Work

Standards labeling matches item text to a code. MathFish and related classifiers ask whether a labeler agrees with a publisher [Lucy et al. 2024]. Agreement between labelers is not a test of whether the tagged operation is required to pass. We keep the publisher’s tag and search for programs that pass without it.

Shortcut learning and partial-input baselines [Geirhos et al. 2020; Gururangan et al. 2018; Kaushik and Lipton 2018; McCoy et al. 2019; Poliak et al. 2018] show that models and rules can answer from the wrong slice of the input. Visual question answering (VQA) models can keep the same answer across images [Agrawal et al. 2016]. Choices-only language models already beat majority on several multiple-choice benchmarks; that success can be a surface artifact or abductive reconstruction of the question [Balepur et al. 2024]. This census is that construction on tagged public mathematics items, not the first choices-only result: the scorer sees the option set, and in the masked condition the question type, but not the given quantities, and we Holm-correct within a

census that includes TIMSS cells. Later papers in the same line separate ranking artifacts from question inference and survey how multiple-choice evaluation is scored [Balepur and Rudinger 2024; Balepur et al. 2025]. Contamination of released items in training data is a rival explanation of those hits [Magar and Schwartz 2022; Sainz et al. 2023]; the Algebra I and Algebra II readers we claim have negative contamination probes. Item-writing and test-wiseness catalogs [Haladyna et al. 2002; Haladyna and Rodriguez 2013; Millman et al. 1965] list surface cues that can determine a choice without content, including using the numbers given without binding them. Some option sets already contain information about which member is the key even with the stem withheld [White and Zammarelli 1981]. Arithmetic closure is that unbound-execution move as a program. Those catalogs are our channels, not claims about students. Test makers and takers hide and seek keys in middle letter positions [Attali and Bar-Hillel 2003]; the lower-central program here picks a middle numeric rank, which is not that position cue. Plug-in substitution of numbered options is a documented SAT mathematics strategy [Katz et al. 2000]; the image-channel row is that strategy as a program on a displayed equation, not a student protocol.

Q-matrix and cognitive-diagnosis methods estimate which skills an item requires from student response data [de la Torre and Chiu 2016; Tatsuoaka 1983]. Expert attribute tags on TIMSS 2011 items have been treated as misspecifiable Q-matrix entries under the generalized deterministic inputs, noisy “and” gate (G-DINA) model [Terzi and Sen 2019]. Differential item functioning (DIF) uses examinee groups to detect items that do not measure the same construct across populations [Holland and Thayer 1986]. This census is the student-free adversarial counterpart: no response vector is used. Comparisons to percent correct, when we make them, use only the statistics TIMSS already published with the public-release items.

Validity, cognitive diagnosis, test-wiseness, choices-only evaluation, and contamination detection each already study one piece of what a tagged correct answer can be taken to mean. Argument-based validity treats a tagged pass as a claim that must be supported for the intended use [American Educational Research Association et al. 2014; Kane 1992, 2013; Messick 1995]. Cognitive diagnosis and differential item functioning estimate attributes or bias from student responses. Test-wiseness catalogs list cues that can determine a choice without the tagged skill. Choices-only and contamination papers measure models on mostly untagged general-knowledge items. Taken together they still do not score this paper’s object: a one-sided program witness on tagged public mathematics items, meaning a program that lacks the tagged operation, never sees the published key, and still matches under the published scoring rule. A match is evidence that a pass does not require the tag. Failure to find a program does not certify the tag. Each family of programs is

Holm-corrected on its own. That object is not a claim about whether students learned.

### 3 Method

#### 3.1 Claims and corpora

Each claim is quoted from its authority. Utah Core grades 6–8 use Utah State Board of Education Core Guides. CCSS seed codes are checked against the same wording. TIMSS uses the 2011 framework definitions of knowing, applying, and reasoning [International Association for the Evaluation of Educational Achievement 2009].

Targets are released summative items and end-of-unit assessments with scoring rules. Practice problems are never targets. This census contains 2,405 keyed public items (2,245 selected-response, 160 numeric): New York Regents 1,448; State of Texas Assessments of Academic Readiness (STAAR) 445; Education Quality and Accountability Office (EQAO) 149; TIMSS 2011 129; earlier TIMSS public-release 106; Programme for International Student Assessment (PISA) 70; New York State Education Department (NYSED) 27; National Assessment Program Literacy and Numeracy (NAPLAN) 21; Massachusetts Comprehensive Assessment System (MCAS) 10. Operational Utah RISE items are secure and are not used. TIMSS 2015 and later are not used.

#### 3.2 Channels

A channel fixes the information and the instruction set, and therefore the operations the program lacks. This census scores three families. The catalog family is the (rule, cell) list locked in the protocol draft; strategy programs were hashed before any pass rate was computed; the language-model family was scored the same day and Holm-corrected inside that family only. Hashes, timestamps, model revisions, seeds, and filter counts are in Appendix A.

**Catalog rules.** Item-writing cues (option length, position, absolute terms, and a signed variant of the catalog numeric-central-value cue) lack every content operation. Stem–option overlap and stem-number copying are shallow partial-input rules. Unbound execution applies a named formula (percent of, unit rate,  $y = kx$ , and the others in the 15-rule catalog) to stem numerals without binding symbols to the situation; it lacks bind and explain. Instruction-only (worked examples from the same unit) is specified but was not run: curriculum scoring keys were not in the public files we could ingest.

**Adaptive strategy programs.** Five programs, written from the same catalogs, were hashed and then scored; later parsing edits did not overwrite version-1 rates. A distractor-derivation hub picks the option from which the most other options can be derived under a fixed set of error transforms. Constraint elimination removes options that a shallow stem keyword rules out (a negative count, a probability outside  $[0, 1]$ , a mean outside the listed data). Arithmetic closure

applies  $+$ ,  $-$ ,  $\times$ ,  $\div$ , squares, and square roots to the stem numbers to depth two and picks the option that first appears. Backsolving parses a one- or two-variable equation from the stem and picks the unique option that satisfies it. A fixed-order portfolio runs those four. None of the five binds a symbol to the situation the standard names. Chance for a rule that eliminates options is the mean of one divided by the number of remaining options, not 0.25. Hub and portfolio must also exceed the 95th percentile of random programs of matched complexity. Cells are every authority-by-claim pair with at least 10 selected-response items. A (program, cell) row enters the family when  $n$  scored is at least 10 and chance is positive: 67 tests. Holm is inside this family. A 172-test pool with the 105 catalog tests is a sensitivity, not a pool with the language-model tests. The catalog unbound formulas already scored in the 105 are a comparison, not this family’s closure program. Text-layer backsolving enters that family only when a cell has  $n \geq 10$  unique fires; it does not. A later addendum, hashed before scoring and not a member of those 67 tests, uses a vision-language model (VLM) to transcribe the displayed equation from a page image, then substitutes each option. The model is not asked to solve. Invented formulas and extracts that do not match the page are abstentions. The addendum bar is  $n$  fired at least 10 and a lower 95% interval strictly above chance. Tie-breaks, the transform set, and the frozen hashes are in Appendix A.1.

**Partial-input language models.** A local instruction-tuned language model sees a slice of the item and picks the option letter with the highest next-token log-probability. There is no chain of thought and no free generation. Two model families, Qwen2.5 (7B-Instruct and 14B-Instruct) and Phi-4, are capable scorers; Mistral-7B-Instruct-v0.3 is a weak-solver check. Two prompts: options only (stem withheld entirely) and masked stem plus options. Masking replaces every numeral, number word, fraction, percentage, decimal, and value binding in the stem with the placeholder  $[N]$ ; question wording, units, operators, and the option strings are kept. Items on which the masker withheld nothing are excluded from the primary population, because the tagged operation may still be performable; the primary population is the 1,215 items with at least one withheld quantity. The witness bar for this channel is:  $n$  at least 10, and the lower 95% bootstrap confidence interval strictly above both format chance and that cell’s modal-letter frequency, on the withheld-quantity population. A pass on those items does not require processing the givens. Contamination probes, not extra family tests, are mean token log-probability of the original unmasked stem plus options, administration-year buckets, option-letter permutation, and a second masking pass that also hides named figure types, variable bindings, and leftover coefficients. Holm for this channel is every scored (model, cell) on the withheld-quantity masked-stem population with  $n$  at least 10: 68 tests (four models, 17 cells). Options-only

is a ladder rung, not a member of that family. Holm is inside this family only. Catalog and strategy stay 105 and 67; the 172-test programmatic pool is a sensitivity and is not mixed with these 68. Model revisions, the letter-token rule, the clean-option filter, and the masking-rule hashes are in Appendix A.2.

### 3.3 Admission and the witness bar

Two kinds of catalog program are admitted.

1. *A priori rules*, taken from the catalogs with a citation each and no fitted parameters, scored on every item they apply to. There are 15 such programs (option length, position, absolute terms, a signed variant of the catalog numeric-central-value cue named below, stem-option overlap, stem-number copying, and unbound formulas).
2. *Fitted or searched programs*, trained on items disjoint from those they are scored on. Headline form is cross-corpus: fit on state items, score on TIMSS, or the reverse. Reported accuracy is held-out only.

Adaptive strategy programs and the language-model channel are separate families, pre-registered on the scoring day as above, not members of the 105. They do not replace the catalog family.

The catalogs list a numeric-central-value cue: among numeric options, the key is often a middle value [Haladyna et al. 2002; Millman et al. 1965]. The *a priori* program we score is a signed variant of that cue, which we call lower-central: parse a leading signed number from each option; if at least three parse, sort them and pick the option at index  $\lfloor (n-1)/2 \rfloor$ . For four options that is the second-smallest number. The catalog median of four numbers is the average of the two central values, which is not an option, so a discrete program must pick one side. That index is fixed in the program; it is not a parameter fit on these pass rates. A protocol draft records option numeric rank (minimum, median, maximum) as searched features, not this exact *a priori* index. The same index is the named tie-break inside the strategy programs.

A catalog or strategy program is a witness on a claim only if  $n$  scored is at least 10, the lower 95% bootstrap confidence interval of the pass rate exceeds format chance (one divided by the number of options on that item, or the mean of one divided by remaining options after elimination; 0.25 on four-option selected items with no elimination), and (for searched, fitted, hub, and portfolio programs) the pass rate is at or above the 95th percentile of random programs of matched complexity, which we call the machine bar. The partial-input language-model bar is stated in the paragraph above and is not loosened. A singleton match is logged, not claimed. Random-weight programs are the baseline, never witnesses.

The searched class is a library of surface-cue programs enumerated to bounded complexity, plus greedy and evolutionary search (Appendix A.3). Fourteen selected on the training corpus are scored on the other corpus only.

### 3.4 Controls

Before a census null means anything, the same searcher has to find witnesses where they are known to exist. We run a synthetic Haladyna-flaw set (circular: the planted cues are the features) and, as the primary control, public multiple-choice corpora whose authors already published an options-only or question-ignored number [Lai et al. 2017; Mihaylov et al. 2018; Talmor et al. 2019; Zellers et al. 2018].

## 4 Results

### 4.1 Controls

On 1,200 synthetic four-option items, each planted atomic program scored 50/50 on holdout, confidence interval  $[1, 1]$ . Recall of planted flaw types is 6/6. False-positive rate on matched clean holdout is 0. That shows the searcher fires when a unique surface cue is planted. It does not show that real released items contain such cues.

On public cue sets, the same 1,847-program space, fit on a train split and scored held-out, is summarized in Table 1. The searcher recovers documented shallow cues on at least two corpora whose authors already published an options-only or question-ignored number. A TIMSS null is therefore not a failure of the tool to find anything.

### 4.2 Power

A claim cannot clear the bar below  $n = 10$ . For larger  $n$ , Table 2 records two floors. The independent-items floor is the smallest observed pass rate whose item-level bootstrap interval excludes chance, at the cell's tagged-item count. The clustered floor is the bypass rate with 80% power to clear that bar when administrations are resampled. Twenty-one cells in this census have  $n \geq 10$ . The smallest independent-items floor is 0.284, on New York Regents Algebra I ( $n = 598$ ); that cell's clustered floor is 0.303. Public-release TIMSS items tagged knowing, applying, or reasoning pool the 2011 cycle with earlier released cycles at  $n = 81/68/86$  tagged items and  $n = 65/54/68$  selected-response items. Independent-item floors on the tagged counts are 0.358/0.353/0.372. Clustered 80% floors on the selected-response counts, from administration resampling in exports/addendum/power-by-cell.json, are 0.415/0.426/0.441. Those TIMSS cells are a null at  $n$  of about 50 to 80.

### 4.3 Census

We ran 350,535 local trials. Fitted stem  $n$ -gram and overlap programs, state to TIMSS and the reverse, do not beat chance on holdout (state to TIMSS applying and reasoning: 36/122 =

**Table 1.** Method control: the same 1,847-program searcher recovers published options-only or question-ignored rates on OpenBookQA and SWAG. Chance is 0.25 except CommonsenseQA (0.20, five options). RACE and CommonsenseQA do not yield a witness here.

| Corpus                             | Published cue number                           | Recovered              | $n$ | Witness         |
|------------------------------------|------------------------------------------------|------------------------|-----|-----------------|
| OpenBookQA                         | 0.496, question ignored [Mihaylov et al. 2018] | 0.450 (longest option) | 329 | yes             |
| SWAG                               | 0.436, ending-only [Zellers et al. 2018]       | 0.565 (searched)       | 62  | yes             |
| RACE [Lai et al. 2017]             | (no official options-only number)              | 0.375                  | 8   | no ( $n < 10$ ) |
| CommonsenseQA [Talmor et al. 2019] | chance 0.20                                    | 0.227                  | 22  | no              |

**Table 2.** Minimum detectable bypass rate at this cell’s  $n$  (every tagged item, not the subset a given rule scores). Independent is an item-level bootstrap on that  $n$ . Clustered is 80% power under administration resampling. TIMSS clustered entries are that resampling on selected-response items ( $n = 65/54/68$  for knowing, applying, reasoning), not a rescaling of the independent floors. TIMSS rows pool 2011 and earlier public-release cycles.

| Authority  | Claim      | $n$ | Independent | Clustered |
|------------|------------|-----|-------------|-----------|
| NY Regents | Algebra I  | 598 | 0.284       | 0.303     |
| NY Regents | Algebra II | 442 | 0.290       | 0.308     |
| NY Regents | geometry   | 408 | 0.294       | 0.309     |
| TIMSS      | applying   | 68  | 0.353       | 0.426     |
| TIMSS      | knowing    | 81  | 0.358       | 0.415     |
| TIMSS      | reasoning  | 86  | 0.372       | 0.441     |
| EQAO       | grade 6    | 53  | 0.396       | 0.415     |

0.295, interval [0.213, 0.377], chance 0.254). Unbound unit-rate,  $y = kx$ , and percent-of do not clear the bar on TIMSS, PISA, or NAPLAN numeric items.

No searched or a priori program cleared the bar on TIMSS knowing, applying, or reasoning. Failure to find a witness on public selected-response  $n$  of 65, 54, and 68 (clustered floors 0.415, 0.426, and 0.441; independent-item floors 0.353 to 0.372) does not certify those tags. It shows that this program space, on these public items, did not refute them.

Fifteen catalog rules were scored on every authority-by-claim cell. The a priori family is every (rule, cell) pair with  $n$  scored at least 10 and a positive format chance: 105 tests. One PISA numeric unit-rate row with chance 0 was excluded as degenerate. Five of those 105 tests clear the pre-registered per-cell bar. None of the five survives Holm or Bonferroni at  $\alpha = 0.05$  (Bonferroni cut 0.00048). There is no family-wise a priori discovery. Two of the five remain witnesses at the per-cell bar and are candidates for replication, not family-wise discoveries (Table 3). Both are the lower-central signed variant of the catalog numeric-central-value cue in Section 3. Their one-sided binomial  $p$  values are 0.0038 (geometry, 90/279) and 0.00070 (EQAO grade 6, 20/40).

The claimed  $n$  is the number of items on which the rule fires (at least three parseable option numbers): 279 of 408

New York Regents geometry items and 40 of 53 EQAO grade 6 items. The remaining items in those cells are not scored by this rule.

On geometry, 19 of 27 administrations have rate above chance (sign test  $p = 0.014$ ); only one administration clears the item bar on its own, so single forms are underpowered and the claim remains the pooled cell. On EQAO grade 6, all three years are above chance by rate; one year clears individually. The  $k = 3$  clustered interval is tighter than the item interval and is descriptive; the existence claim is the item interval on  $n = 40$ , not a stable cell law of EQAO grade 6.

On geometry the cue is central-ness: upper-central 0.330 ( $n = 279$ , item interval [0.272, 0.387]) and nearest-to-mean 0.312 also clear; smallest and largest both 0.219 do not. On EQAO grade 6 only the pre-registered lower-central index clears; upper-central is 0.250 (item interval [0.125, 0.400]). The EQAO witness is specific to the second-smallest option, not to a middle option in general.

The one-sided claim covers only the matched subset: the 90 geometry items and 20 EQAO grade 6 items on which the rule returns the published key. On those items, matching the key does not require the tagged operation. The rule is a cell-level witness at the per-cell bar on the 279 and 40 items it fires on. It is not a witness against each unmatched item in the cell.

The other three uncorrected clears are not claimed. New York Regents Algebra II, same lower-central program, is 0.298 at  $n = 372$ . The item interval is [0.253, 0.344] and the clustered interval is [0.255, 0.341]; the lower end sits on chance. Texas Essential Knowledge and Skills (TEKS) Algebra I stem-number repeat: 0.395,  $n = 43$ , interval [0.256, 0.535]. TEKS grade 5 unit-rate: 0.417,  $n = 12$ , interval [0.167, 0.750], chance 0.146. Searched programs that clear the machine bar on TEKS grade 6 and on Regents Algebra I / geometry have  $n$  between 11 and 20. We do not claim them: that  $n$  is the witness floor, 1,847 programs were searched, and all four searched programs fail a late-administration held-out test (late  $n = 6$  to 10, intervals include chance). Fitted  $n$ -grams fail every temporal slice (state-pooled 0.257,  $n = 1243$ ). One STAAR refit reaches 0.643 on  $n = 14$  and is not claimed (train  $n = 8$ , one form, 1,562 programs searched).

**Table 3.** Per-cell catalog existence rows: they beat chance in that cell and are candidates for replication, not family-wise discoveries.  $n$  is items the rule fires on, not the cell size in Table 2. Item interval resamples items; clustered interval resamples administrations (year-session for Regents, year for EQAO). Neither row survives Holm (family-wise multiple-testing adjustment) over 105 a priori tests. The EQAO row is specific to the second-smallest option; the existence claim is the item interval on  $n = 40$ .

| Authority  | Claim    | $n$ | Pass  | Item 95% CI    | Clustered 95% CI |
|------------|----------|-----|-------|----------------|------------------|
| NY Regents | geometry | 279 | 0.323 | [0.265, 0.380] | [0.265, 0.374]   |
| EQAO       | grade 6  | 40  | 0.500 | [0.350, 0.650] | [0.417, 0.588]   |

Table 4 is every cell on which the claimed rule scored  $n \geq 10$ . Two cells clear the per-cell bar and are claimed. Algebra II clears that bar and is not claimed. Over this 17-cell one-rule slice, Holm at  $\alpha = 0.05$  keeps EQAO grade 6 and drops geometry. That slice is exploratory: it is the cells this one rule scored, defined after seeing which rule cleared, and the protocol names no primary family. TIMSS reasoning is 0.368,  $n = 57$ , interval [0.246, 0.491]: the interval includes chance, so it is not a witness.

Four further catalog cues were added after the primary scoring: longest option, convergence, stem-word overlap, and mean of other options. Hedge and grammatical never uniquely applied at  $n \geq 10$ . They give 22 further tests (127 total). The primary Holm family remains the 105 pre-registered tests. The combined family still has no Holm survivor. One further per-cell clear, not independent of the central-value cue, is mean of other options on geometry, 0.327 ( $n = 156$ , interval [0.256, 0.404]).

#### 4.4 Adaptive strategy channels

The five strategy programs in Section 3 are a second family: 67 tests, hashed before scoring, Holm inside the family. Three rows clear the per-cell bar. A 172-test pool with the 105 catalog tests is a sensitivity: the same single Holm reject, and none of the original 105 become Holm rejects. Catalog, strategy, and language-model tests are three Holm families, never one mixed pool.

**Arithmetic closure (unbound execution).** On STAAR grade 5 the depth-2 closure fires on 40 of 60 selected items and matches 21 of those 40 keys. The rate is 0.525 ( $n = 40$ , item 95% CI [0.375, 0.675], chance 0.250, one-sided binomial  $p = 0.000175$ ). That  $p$  is the smallest in the 67-test family (Holm threshold  $0.05/67 = 0.000746$ ) and remains the only Holm reject in that family and in the 172-test programmatic sensitivity (threshold  $0.05/172 = 0.000291$ ). It is a discovery-set Holm reject on the frozen 2019/2021/2022 census. It did not replicate on held-out administrations, below. The catalog unit-rate row on the same cell (0.417,  $n = 12$ , chance 0.146) was already scored and unclaimed; closure is a different program with larger  $n$ , not a second independent mechanism. Grade 5 is the only closure cell that clears on the discovery set.

Geometry is 0.196 ( $n = 224$ , interval [0.143, 0.250]), which does not clear.

The registered rule still holds under the pre-registered robustness cuts: depth 1 only is  $17/30 = 0.567$  [0.400, 0.733]; depth 2 without the geometry extras  $\pi$  and  $ab/2$  is  $20/40 = 0.500$  [0.350, 0.650]. Lower-central alone on the same 40 items is  $13/40 = 0.325$  [0.200, 0.475] and does not clear. Random tie-breaks among minimum-depth options give expected rate 0.367 with a 1,000-draw interval [0.250, 0.475]; observed 0.525 sits above that interval. Twenty-seven of the 40 picks were settled by the lower-central tie-break among minimum-depth options (15 of those 27 correct); 13 were settled by depth alone. Mean closure density is 3.3 of 4 options. Clustered by STAAR year the interval is [0.455, 0.615]: 2019 8/13, 2021 8/16, 2022 5/11. The sign test on three administrations above chance is  $p = 0.125$ . Those three forms are the whole cell.

Thirteen of the 40 fired stems contain artifact numerals (axis ticks, thousands separators, stem-and-leaf digits, footer grade numbers, leaked neighboring text). That is a construct problem for those 13. The remaining 27, a population defined after the fact, are  $16/27 = 0.593$  [0.370, 0.778]. Unique depth-1 closure (exactly one option in the depth-1 set) is 10 of 40 fired, and that unique option is the key on 6 of those 10;  $n = 10$  is the bar edge. The frozen census row remains the Holm reject in the 67-test discovery family. It is not a confirmed family-wise discovery.

**Replication on held-out STAAR grade 5.** The registered prediction was that filtered arithmetic closure on new grade 5 items alone would have a lower 95% CI strictly above 0.250. On the version-1 unrepaired parse, 18 of 44 keys match (0.409, CI [0.273, 0.545]): the prediction is met, and  $n = 44$  is below the registered power floor of 46. That positive is not described as underpowered. After a mechanical last-option page-leak repair, hashed before rescoring (42 last-option repairs, 3 stacked-fraction rejoins; hand-check 14 agree / 4 disagree / 2 uncertain, 0 key mismatches), 17 of 44 match (0.386, CI [0.250, 0.545]). The prediction is not met. That null is underpowered ( $n = 44 < 46$ ). English grade 5 forms for 2017 and 2018 were then parsed (the 2015 key, the 2020 test, and the 2023 test were not found). New forms alone are 4 of 20 (0.200, CI [0.050, 0.400]). Combined repaired plus

**Table 4.** Lower-central on every cell with  $n \geq 10$  (items the rule fires on). Geometry and EQAO grade 6 are per-cell witnesses. Algebra II clears the bar and is not claimed. Holm on this 17-cell slice is exploratory. The other 88 of 105 a priori tests are omitted.

| Authority  | Claim      | $n$ | Pass  | 95% CI         | Chance | Bar               |
|------------|------------|-----|-------|----------------|--------|-------------------|
| NY Regents | geometry   | 279 | 0.323 | [0.265, 0.380] | 0.250  | yes (claimed)     |
| EQAO       | grade 6    | 40  | 0.500 | [0.350, 0.650] | 0.254  | yes (claimed)     |
| NY Regents | Algebra II | 372 | 0.298 | [0.253, 0.344] | 0.250  | yes (not claimed) |
| NY Regents | Algebra I  | 458 | 0.260 | [0.223, 0.301] | 0.250  | no                |
| TIMSS      | reasoning  | 57  | 0.368 | [0.246, 0.491] | 0.249  | no                |
| TIMSS      | applying   | 50  | 0.240 | [0.120, 0.360] | 0.249  | no                |
| TIMSS      | knowing    | 41  | 0.293 | [0.146, 0.439] | 0.248  | no                |
| TEKS       | Algebra I  | 61  | 0.213 | [0.115, 0.328] | 0.250  | no                |
| TEKS       | grade 6    | 52  | 0.327 | [0.192, 0.462] | 0.250  | no                |
| TEKS       | grade 5    | 49  | 0.347 | [0.224, 0.490] | 0.250  | no                |
| TEKS       | grade 8    | 45  | 0.267 | [0.156, 0.400] | 0.250  | no                |
| TEKS       | grade 7    | 41  | 0.317 | [0.171, 0.463] | 0.250  | no                |
| TEKS       | grade 3    | 36  | 0.278 | [0.139, 0.444] | 0.250  | no                |
| TEKS       | grade 4    | 35  | 0.314 | [0.171, 0.486] | 0.250  | no                |
| EQAO       | grade 9    | 38  | 0.211 | [0.079, 0.342] | 0.257  | no                |
| EQAO       | grade 3    | 33  | 0.242 | [0.091, 0.394] | 0.253  | no                |
| NAPLAN     | year 7     | 14  | 0.214 | [0.000, 0.429] | 0.250  | no                |

new is 21 of 64 (0.328, CI [0.219, 0.438]). Here  $n = 64 \geq 46$ . The prediction is not met, and this null is not underpowered. Grade 4 lower-central is still 10 of 17 after repair; 5 of those 10 hits are parse artifacts, and the clean 5 of 9 does not clear. It is not a witness. The Holm-surviving unbound-execution row on the frozen STAAR 2019/2021/2022 grade 5 cell did not replicate on held-out 2013/2014/2016/2017/2018 administrations after the pre-registered option-leak repair. Discovery 0.525, combined replication 0.328 with an interval that covers chance.

**Hub, elimination, portfolio.** The distractor-derivation hub never clears once the random-program 95th percentile is required. Unique-hub share among 992 family-scored items is 0.189; most hubs are ties broken by lower-central. Geometry is 0.255 ( $n = 161$ , interval [0.193, 0.317]) versus chance 0.250 and p95 0.311. EQAO grade 6 is 0.321 ( $n = 28$ ) versus p95 0.438. TEKS grade 4 is the only chance-only clear (0.455,  $n = 22$ , interval [0.273, 0.682] versus p95 0.476) and is not a witness.

Constraint elimination barely eliminates: mean remaining options stay near 4 (geometry 3.92, EQAO grade 6 3.77), and the share of options removed is 0.025 on geometry and 0.019 on EQAO grade 6. Two specified intersections kill the published key (probability with percent-point options; interior-angle (0, 180) on a rectangle angle-sum). Key-survival is 0.985 on geometry, 0.906 on EQAO grade 6, and 0.877 on TEKS grade 8. A hand check of 15 keyword-matched items keeps 13 of 15 keys. The two per-cell clears, geometry 0.307 ( $n = 404$ , interval [0.262, 0.351] versus effective chance 0.254) and EQAO grade 6 0.412 ( $n = 51$ , interval [0.275, 0.529] versus 0.268), are the lower-central rule again. The keyword-matched geometry subset still clears (0.352,  $n = 162$ ); the EQAO keyword subset ( $n = 19$ ) does not. We keep the catalog lower-central rows as the claimed witnesses on those

cells. TIMSS reasoning is 0.379 ( $n = 66$ , interval [0.258, 0.500] versus effective chance 0.263) and does not clear.

The portfolio never clears the random-program 95th percentile (Table 5). EQAO grade 6 clears chance and one-over- $k$  (0.453,  $n = 53$ , interval [0.321, 0.585] versus 0.253) and still fails p95 0.472. Almost all mass is lower-central among survivors.

**Image-channel back-solving.** Text-layer back-solving still does not produce a 67-family row (6 unique fires among 1,186 solving-tagged stems; no cell  $n \geq 10$ ). Most equations in the source PDFs are images. The image addendum was scored after that family and is not folded into those 67 Holm tests. Unique-satisfier fired on 17 of 598 Algebra I solving-tagged selected-response items; two fires are recoded as abstentions (an invented formula with no displayed equation; an incomplete two-equation extract). On the remaining 15, 14 match the key (0.933,  $n = 15$ , item 95% CI [0.800, 1.000], chance 0.25). Coverage is  $15/598 = 0.025$ . The miss is a correctly transcribed equation on an item that asks which point is not on the graph. No other named cell reaches  $n \geq 10$  fired. The row is an existence measurement after the catalog, like the geometry and EQAO lower-central rows, not a family-wise discovery and not a census of Algebra I. On those 14 matched items a pass does not require solving.

**Strategy controls.** On the same OpenBookQA and SWAG hold-out splits used in Table 1, elimination and portfolio sit on chance (OpenBookQA elimination 0.260,  $n = 400$ , interval [0.218, 0.302]; portfolio 0.268 [0.225, 0.310]; SWAG elimination 0.248 [0.205, 0.288]; portfolio 0.240 [0.198, 0.280]). Numeric hub, closure, and back-solve do not fire on those language items.

**Table 5.** Fixed-order strategy portfolio on the five headline cells. Rate is against effective chance of the branch used. Witnesses in this channel must also exceed the 95th percentile of 200 random programs of matched complexity. No cell clears that bar.

| Authority  | Claim     | $n$ | Pass  | Eff. chance | 95% CI         | p95   | Bar |
|------------|-----------|-----|-------|-------------|----------------|-------|-----|
| NY Regents | geometry  | 408 | 0.279 | 0.253       | [0.238, 0.321] | 0.297 | no  |
| EQAO       | grade 6   | 53  | 0.453 | 0.267       | [0.321, 0.585] | 0.472 | no  |
| TIMSS      | knowing   | 65  | 0.308 | 0.271       | [0.200, 0.431] | 0.292 | no  |
| TIMSS      | applying  | 54  | 0.333 | 0.254       | [0.204, 0.463] | 0.370 | no  |
| TIMSS      | reasoning | 68  | 0.309 | 0.263       | [0.206, 0.426] | 0.338 | no  |

#### 4.5 Partial-input language-model channel

A mechanical clean-option filter keeps 1,496 of 2,245 selected-response items, dropping exam footers, STAAR item codes, country-table fragments, options longer than 120 characters, and bleed from the next stem. TIMSS 2011 is mostly removed by country-table contamination (17 kept). Scorers take the letter with the highest next-token log-probability, with no chain of thought. The witness bar for this channel is  $n \geq 10$  and a lower 95% interval strictly above both format chance and the cell's modal-letter frequency, on items where at least one quantity was withheld. A pass on those items does not require processing the givens. It is not a claim that the model reasons, and not a claim that students guess. One rater on the pre-registered 40-item Algebra I masked-stem sheet scored  $15/40 = 0.375$  (bootstrap 95% CI [0.225, 0.525], no items marked recognized). The lower bound is not strictly above chance 0.250 or the cell modal frequency 0.271; clearing at  $n = 40$  needs  $k = 17$ . That is an uninformative null, not a human witness. Qwen 14B on the same 40 is 0.425; on the 28 items whose census row matches the printed packet, Cohen's  $\kappa$  is 0.212 (human 9/28, 14B 14/28). The Algebra I and Algebra II language-model claim is a model fact and is unchanged.

**Options only.** On the 1,487 scoreable clean items, both local models score 0.301 versus chance 0.250 and always-B 0.275 (7B interval [0.277, 0.326]; 14B [0.276, 0.323]). Letter agreement between models is 0.457. A with-stem sanity check on 7B is 0.460 on 200 items. A proprietary assistant scored through an audited no-tool transcript scores 0.359 ( $n = 1493$ , interval [0.334, 0.384]). No  $n \geq 10$  cell has a 7B lower interval above that cell's modal-letter frequency. Algebra I is 0.309 [0.265, 0.355] versus modal B 0.272 (111 of 408). On the 1,026 items whose key is not the cell modal letter, 7B is 0.318 [0.288, 0.347] versus chance 0.250. On 14B, Algebra I is 0.321 [0.279, 0.363] versus modal 0.272: the lower interval exceeds the modal frequency. That options-only Algebra I row is not a claimed witness; the claimed object is the masked-stem condition below. Recency on the proprietary scorer minus 7B is largest before 2015 (+0.112,  $n = 178$ ), then +0.024 in 2015–2019 ( $n = 578$ ) and +0.069 in 2020 and later ( $n = 712$ ).

**Masked stem plus options.** Every numeral, number word, fraction, percentage, decimal, and value binding in the stem is replaced by  $[N]$ . Question wording, units, and options remain. Of 1,487 scored items, 272 had nothing withheld; the primary population is the other 1,215. First-run pooled rates on all 1,487 (7B 0.405, 14B 0.425) mix those 272 in and are not the claim. On the withheld-quantity population, Regents Algebra I is  $139/358 = 0.388$  [0.341, 0.439] on 7B and  $149/358 = 0.416$  [0.363, 0.469] on 14B, versus chance 0.250 and modal B 0.271 (97 of 358). Algebra II is  $91/248 = 0.367$  [0.306, 0.423] on 7B and  $89/248 = 0.359$  [0.298, 0.415] on 14B, versus chance 0.250 and modal C 0.278 (69 of 248). Both cells clear both bars on both models (Table 7). Geometry does not beat its modal letter 0.289 on 7B (0.319,  $n = 166$ , interval [0.253, 0.392]) and does not clear chance on 14B (0.301 [0.235, 0.373]). EQAO grade 6 does not clear chance on 7B (0.400,  $n = 40$ , interval [0.225, 0.575] versus 0.255). TIMSS knowing on this population does not beat its modal letter 0.409 on 7B (0.500,  $n = 22$ , interval [0.318, 0.682]). The first-run TIMSS knowing rate 0.657 ( $n = 35$ ) was carried by items where nothing was withheld: of 35 items, 18 are knowledge-answerable without any quantity, 12 are cue-readable from the leftover option set, and 5 are unclear.

One-sided binomial  $p$  against chance on the withheld-quantity population: Algebra I  $5.7 \times 10^{-9}$  (Qwen 7B),  $4.5 \times 10^{-12}$  (Qwen 14B), and  $9.4 \times 10^{-13}$  (Phi-4); Algebra II  $2.9 \times 10^{-5}$  (7B),  $9.0 \times 10^{-5}$  (14B), and  $4.2 \times 10^{-4}$  (Phi-4); Phi-4 geometry  $9.9 \times 10^{-6}$  (67/166). Against the cell modal frequency: Algebra I  $9.4 \times 10^{-7}$ ,  $2.2 \times 10^{-9}$ , and  $5.6 \times 10^{-10}$ ; Algebra II 0.00147, 0.00343, and 0.011; Phi-4 geometry 0.00104. The language-model Holm family is every scored (model, cell) on the withheld-quantity masked-stem population with  $n \geq 10$ : 68 tests (four models, 17 cells). Options-only is a ladder rung, not a member of that family. Holm at  $\alpha = 0.05$  in that family, against chance, rejects both Qwen Algebra I rows, both Qwen Algebra II rows, Phi-4 Algebra I, Phi-4 Algebra II, Phi-4 geometry, and Qwen 7B TEKS grade 8. Claimed cells are those that clear the registered channel bar, lower 95% CI above chance and above the cell modal-letter frequency, on a capable scorer. Scorers are not averaged. That set is Algebra I and Algebra II on Qwen 7B, Qwen 14B, and Phi-4. Geometry on Phi-4 meets the cell bar (67/166 = 0.404, CI [0.331, 0.476], versus chance 0.250 and modal B 0.289) but not

the non-modal-key split (0.229,  $n = 118$ , CI [0.161, 0.305]), which is at or below chance; the cell bar is carried by B-keyed items, and geometry is not claimed. Qwen geometry does not clear the cell bar. Qwen 7B TEKS grade 5 (18/38 = 0.474, CI [0.316, 0.632] versus modal 0.263) and grade 8 (12/18 = 0.667, CI [0.444, 0.889] versus modal 0.389) clear the bar on 7B only and fail it on Qwen 14B and on Phi-4; they are scorer-specific rows, not claimed witnesses. Qwen 14B TIMSS knowing rejects Holm against chance and does not beat its population modal 0.409 (lower CI 0.409). Mistral clears none of these cells on both bars.

**Contamination probes.** On the withheld-quantity items, 7B masked-stem accuracy by quartile of mean token log-probability of the original unmasked stem plus options is 0.349 in Q1 versus 0.439 in Q4 (Spearman 0.091; 14B Spearman 0.107, Q4 minus Q1 0.120). Recency on all 1,487 masked-stem 7B items is flat: 0.416 / 0.409 / 0.399 / 0.402 across before 2015, 2015 to 2019, 2020 to 2023, and 2024 and later ( $n = 178 / 579 / 383 / 331$ ). Option permutation on the withheld-quantity items (7B) is 0.385 [0.357, 0.412] versus 0.376 unpermuted; Algebra I is 0.405 [0.355, 0.453]. Version-2 masking (named figures, bindings, leftover coefficients) on the version-1 withheld-quantity overlap is 0.370 versus 0.376, delta  $-0.007$  (McNemar 50 version-2-only correct, 58 version-1-only). Extra masking does not create the channel.

**Scorer dependence.** The claim is existential. A reader exists in the Qwen2.5 family (7B-Instruct and 14B-Instruct) that recovers the key from question type plus option form on withheld-quantity Algebra I and Algebra II; the contamination probes above are negative for that reader. A weaker reader does not. Mistral-7B-Instruct-v0.3 (revision in Appendix A.2) on the primary population: Algebra I 0.302 [0.257, 0.349] clears chance 0.250 but not modal B 0.271; Algebra II 0.262 [0.210, 0.319] clears neither; pooled masked-stem 0.277 versus options-only 0.249. We report that row in Table 7 and do not average across scorers.

Phi-4, a 14-billion-parameter instruct model outside the Qwen family, clears both bars on Algebra I and Algebra II: Algebra I 0.422 [0.374, 0.472] versus chance 0.250 and modal B 0.271; Algebra II 0.347 [0.286, 0.407] versus chance 0.250 and modal C 0.278. Algebra I residual with none of the four catalog cues is 0.380 [0.307, 0.460] ( $n = 163$ ); non-modal-key accuracy is 0.387 ( $n = 261$ , CI above 0.250). Numeric options also clear chance (0.366,  $n = 71$ ). Geometry meets the cell bar (0.404 [0.331, 0.476] versus chance 0.250 and modal B 0.289,  $n = 166$ ) but not the non-modal-key split (0.229,  $n = 118$ , CI [0.161, 0.305]) and is not claimed. Pooled masked-stem is 0.401 [0.374, 0.426] versus options-only 0.291 (McNemar 277/143). Mistral's with-stem letter-argmax ceiling matches its masked-stem rate: Algebra I 0.304 [0.257, 0.352] (chance only versus modal 0.271) and Algebra II 0.290 [0.230, 0.347] (neither), against masked-stem 0.302 and 0.262. That miss is capability, not a missing cue. The Algebra I and Algebra II

claim remains existential and holds in two model families, Qwen2.5 and Phi-4 (Table 6). Qwen 7B does not beat geometry's modal letter, and Qwen 14B does not clear chance on that cell.

**What the channel measures.** Cue attribution on Qwen 14B Algebra I: rate when the key is versus is not the lower-central option 0.341 ( $n = 82$ ) versus 0.438 ( $n = 276$ ); modal letter 0.454 ( $n = 97$ ) versus 0.402 ( $n = 261$ ); hub 0.370 ( $n = 46$ ) versus 0.423 ( $n = 312$ ); unique longest 0.333 ( $n = 42$ ) versus 0.427 ( $n = 316$ ); residual with none of the four cues 0.442 [0.368, 0.521],  $n = 163$ , versus chance 0.250. Option type: algebraic expressions 0.462 ( $n = 65$ ), text 0.452 ( $n = 188$ ), numeric 0.310 ( $n = 71$ ), mixed 0.385 ( $n = 26$ ), ordered pairs 0.250 ( $n = 8$ ). Phi-4 on the same Algebra I split: residual 0.380 [0.307, 0.460] ( $n = 163$ ); algebraic 0.369 ( $n = 65$ ), text 0.473 ( $n = 188$ ), numeric 0.366 [0.254, 0.479] ( $n = 71$ , clears chance), mixed 0.308 ( $n = 26$ ), ordered pairs 0.500 ( $n = 8$ ). On Qwen 14B the leftover is strongest on expression and text options and is weak on numeric options; Phi-4 also clears chance on numeric options. The reader uses cues outside the catalog. On those items a pass does not require executing the tagged computation on the givens; it may require recognizing the form the wording names.

**An item with the quantities hidden.** New York Regents Algebra I, January 2023, item 22.

The item id is nyregents-algebra-i-2023-jan-q22.

Masked stem:

If the zeros of the function  $g(x)$  are  $\{[N], [N], [N]\}$ , which function could represent  $g(x)$ ?

- A.  $g(x) = (x + 3)(x - 4)$
- B.  $g(x) = (x - 3)(x + 4)$
- C.  $g(x) = x(x + 3)(x - 4)$
- D.  $g(x) = x(x - 3)(x + 4)$

Published key: C. Qwen 14B letter-argmax picks C. The visible cue is the three placeholders in the zero set, which match the three-factor option that includes an  $x$ . That is form recognition of  $n$  factors from  $n$  placeholders, not execution of the tagged computation on the given zeros.

On the same withheld-quantity Algebra I and Algebra II items, letter-argmax with the original stem (no chain of thought) is higher than masked-stem, and masked-stem is higher than options-only (Table 8).

#### 4.6 An item the lower-central rule answers

EQAO 2023 grade 6, released questions, printed card page 8 (key table page 21). Stem: "Which number has 4 as its tenths digit?"

- A. 92.174   B. 85.043   C. 79.451   D. 43.256

Published key: C. File tag B1-KU: B1 is EQAO's content code on this item; KU is the knowledge-and-understanding category. Item id eqao-2023-g6-q1.

**Table 6.** With-stem letter-argmax ceiling versus masked-stem (given quantities replaced by placeholders) on withheld-quantity Algebra I ( $n = 358$ ). Bar: lower 95% CI above chance 0.250 and modal B 0.271. Mistral’s ceiling is the masked-stem rate.

| Scorer     | With stem            | Masked stem          | Bar         |
|------------|----------------------|----------------------|-------------|
| Mistral 7B | 0.304 [0.257, 0.352] | 0.302 [0.257, 0.349] | chance only |
| Qwen 7B    | 0.508 [0.461, 0.559] | 0.388 [0.341, 0.439] | both        |
| Qwen 14B   | 0.575 [0.525, 0.626] | 0.416 [0.363, 0.469] | both        |
| Phi-4      | 0.497 [0.447, 0.553] | 0.422 [0.374, 0.472] | both        |

**Table 7.** Partial-input language-model channel on named cells, Qwen and Mistral, withheld-quantity population. Chance is mean  $1/k$ . Modal is the cell’s modal key-letter frequency on that population. Bar:  $n \geq 10$  and lower 95% CI above chance and above that modal frequency. Claimed cells are Algebra I and Algebra II on Qwen2.5-7B and 14B; Phi-4 on those two cells is in Table 6. Geometry is an unlabeled row: Phi-4 meets the cell bar but not the split on items whose key is not the cell’s most common letter (0.229,  $n = 118$ , CI [0.161, 0.305]) and is not claimed. TEKS grade 5 and grade 8 clear on Qwen 7B only and are not claimed. Mistral-7B-Instruct-v0.3 does not clear.

| Claim           | $n$ | Qwen 7B              | Qwen 14B             | Mistral 7B           | Chance | Modal | Bar                  |
|-----------------|-----|----------------------|----------------------|----------------------|--------|-------|----------------------|
| Algebra I       | 358 | 0.388 [0.341, 0.439] | 0.416 [0.363, 0.469] | 0.302 [0.257, 0.349] | 0.250  | 0.271 | Qwen; not Mistral    |
| Algebra II      | 248 | 0.367 [0.306, 0.423] | 0.359 [0.298, 0.415] | 0.262 [0.210, 0.319] | 0.250  | 0.278 | Qwen; not Mistral    |
| geometry        | 166 | 0.319 [0.253, 0.392] | 0.301 [0.235, 0.373] | 0.295 [0.229, 0.373] | 0.250  | 0.289 | none                 |
| EQAO grade 6    | 40  | 0.400 [0.225, 0.575] | 0.425 [0.300, 0.575] | 0.150 [0.050, 0.275] | 0.255  | 0.475 | none                 |
| TIMSS knowing   | 22  | 0.500 [0.318, 0.682] | 0.591 [0.409, 0.773] | 0.364 [0.136, 0.545] | 0.248  | 0.409 | none                 |
| TIMSS applying  | 24  | 0.375 [0.208, 0.583] | 0.375 [0.208, 0.583] | 0.167 [0.042, 0.333] | 0.248  | 0.333 | none                 |
| TIMSS reasoning | 42  | 0.286 [0.167, 0.429] | 0.310 [0.167, 0.452] | 0.238 [0.119, 0.381] | 0.261  | 0.286 | none                 |
| TEKS grade 5    | 38  | 0.474 [0.316, 0.632] | 0.368 [0.211, 0.526] | 0.184 [0.079, 0.316] | 0.250  | 0.263 | 7B only; not claimed |
| TEKS grade 8    | 18  | 0.667 [0.444, 0.889] | 0.611 [0.389, 0.833] | 0.389 [0.167, 0.611] | 0.250  | 0.389 | 7B only; not claimed |

**Table 8.** Information ladder on withheld-quantity Algebra I ( $n = 358$ ) and Algebra II ( $n = 248$ ), Qwen 7B and 14B. Each rung adds one kind of information; the gap between masked-stem and with-stem is what the given quantities buy.

| Claim      | Model    | Chance | Modal                | Options              | Masked               | With stem            |
|------------|----------|--------|----------------------|----------------------|----------------------|----------------------|
| Algebra I  | Qwen 7B  | 0.250  | 0.271 [0.226, 0.318] | 0.313 [0.268, 0.360] | 0.388 [0.341, 0.439] | 0.508 [0.461, 0.559] |
| Algebra I  | Qwen 14B | 0.250  | 0.271 [0.226, 0.318] | 0.332 [0.285, 0.380] | 0.416 [0.363, 0.469] | 0.575 [0.525, 0.626] |
| Algebra II | Qwen 7B  | 0.250  | 0.278 [0.222, 0.331] | 0.274 [0.218, 0.335] | 0.367 [0.306, 0.423] | 0.403 [0.339, 0.464] |
| Algebra II | Qwen 14B | 0.250  | 0.278 [0.222, 0.331] | 0.286 [0.234, 0.343] | 0.359 [0.298, 0.415] | 0.440 [0.379, 0.500] |

The sorted numbers are 43.256, 79.451, 85.043, 92.174. The rule picks index  $\lfloor (4 - 1)/2 \rfloor = 1$ , which is 79.451, option C. The program never reads the stem and does not inspect tenths digits. Matching the key on this item does not require identifying a tenths digit.

The geometry row is the same program on 279 Part I items, the multiple-choice session of the New York Regents geometry examination. Several of our text-layer stems are too messy to print; the rate is the result, not a single photographed page.

A 20-item printed-page audit, 10 geometry lower-central hits and 10 Algebra I masked-stem hits, compared each census row to the source exam PDF (no optical character recognition). Eight rows match stem, options, and key. Eight have footer, figure-label, or previous-item glue on an option or stem. Three are truncated. One Algebra I row has the printed stem and a different option set (June 2023 item 3: printed lines  $y = 2x - 2$  and three neighbors; census inequalities). No sampled PDF was missing. Glue on geometry option D is

usually a footer after a correct leading number. The Algebra I language-model claim is on  $n = 358$ ; one wrong option set in a 10-item Algebra I sample does not remove that cell, and it is a construct caveat on the text layer.

EQAO grade 6 and NY Regents geometry are exam or grade labels, not TIMSS reasoning. We do not rank Ontario, New York, or TIMSS.

#### 4.7 What would refute these claims

The two catalog lower-central claims would be false if the published keys on the scored items were wrong, if the implemented rule picked any index other than  $\lfloor (n - 1)/2 \rfloor$  on the sorted parsed option values, or if a recomputed lower 95% bootstrap confidence interval on the same correct/incorrect flags did not exceed chance, including the administration-clustered interval. A family-wise Holm miss is not a refutation of the per-cell bar; it is reported above. An EQAO witness under upper-central or nearest-to-mean would also change the specification claim on that row.

The STAAR grade 5 closure discovery would be false if the published keys on those 40 fired items were wrong, if the implemented depth-2 closure with the registered tie-break did not produce 21 correct, or if the one-sided binomial  $p$  against chance 0.25 were larger than the Holm threshold on 67. That row is a discovery-set Holm reject that failed replication: the registered prediction on held-out filtered grade 5 items was a lower 95% CI strictly above 0.250, and the powered combined set is  $21/64 = 0.328$ , CI  $[0.219, 0.438]$ . Artifact numerals on 13 discovery stems and the three-administration sign test remain caveats on the discovery construct.

The Algebra I and Algebra II masked-stem claims would be false if a recomputed lower 95% interval on the withheld-quantity items did not exceed both chance and the cell modal-letter frequency on that scorer, or if the masking left a computable given quantity in the stem. A Holm miss against chance in the 68-test family would remove the family-wise statement; it would not by itself remove the per-cell channel bar if the interval still cleared. Mistral failing the same bar does not refute the Qwen or Phi-4 rows. A with-stem ceiling at the masked-stem rate, as on Mistral, is a capability miss, not a missing-cue miss.

The Algebra I image-backsolve existence row would be false if honest  $n$  fired were below 10, if the lower 95% interval on those 15 flags were not strictly above chance 0.25, or if the 14 hits were invented formulas rather than displayed equations. A Holm statement about this row is not claimed.

A TIMSS knowing, applying, or reasoning witness would have been any catalog, strategy, or partial-input program that clears the bar stated for its channel, including the withheld-quantity language-model bar. No such row exists in this file. Public selected-response  $n$  is 65, 54, and 68. Bypass rates below the clustered floors in Table 2 would not have been seen at that  $n$ .

## 5 Limitations

We did not measure students. A witness is a fact about a key and a program. It is not a claim that students use the program.

The one-sided claim covers only the matched subset of each claimed cell: the 90 of 279 New York Regents geometry items and the 20 of 40 EQAO grade 6 items on which the lower-central rule returns the published key; the Algebra I and Algebra II items on which the masked-stem model returns the key; and the 14 of 15 Algebra I displayed-equation items on which unique substitution returns the key. A pass on an item the program does not match has not been shown to omit the tagged operation.

We did not exhaust the class of programs that lack the tagged operation. A later search can find a TIMSS witness this one missed. On the withheld-quantity population, no channel in this file clears TIMSS knowing, applying, or reasoning.

Public selected-response  $n$  is about 50 to 80. That null is not a certification of the tags.

Instruction-only (worked examples from the same unit) is specified in the protocol and was not run: curriculum scoring keys were not in the public files we could ingest.

A 20-item printed-page audit of 10 geometry lower-central hits and 10 Algebra I masked-stem hits found 8 matches of stem, options, and key, 8 glue cases, 3 truncations, and 1 wrong option set (Algebra I June 2023 item 3). Rates in this paper are computed on the text layer. A witness on a corrupted option set does not refute the printed tag.

The EQAO grade 6 catalog witness is specific to the pre-registered lower-central index. Upper-central on that cell is chance.

STAAR grade 5 closure on the frozen 2019/2021/2022 cell is a discovery-set Holm reject ( $21/40 = 0.525$ ) that did not replicate on held-out 2013/2014/2016/2017/2018 administrations after a pre-registered option-leak repair (combined  $21/64 = 0.328$ , CI  $[0.219, 0.438]$ ,  $n \geq 46$ ). Thirteen of 40 discovery stems contain artifact numerals. The artifact-free discovery rate  $16/27$  is on a population defined after the fact.

Text-layer back-solving does not reach  $n \geq 10$  fired on any cell. The image addendum measures it on 15 of 598 Algebra I solving-tagged items (coverage 0.025); that is not a census of Algebra I. The row is not a 67-family Holm test.

The language-model channel has the operations. What is withheld is information: the stem, or every given quantity. The inference is that the published key is determined by question type plus the option set, not that the model solved the tagged exercise and not that a student would match the key from that slice. The model is not a student. Items on which the masker withheld nothing are excluded from that claim; TIMSS knowing's first-run rate was partly those items. The Algebra I and Algebra II witnesses hold on Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, and Phi-4. Geometry on Phi-4 meets the cell bar (0.404,  $n = 166$ , lower CI above chance and above modal B 0.289) but not the non-modal-key split (0.229,  $n = 118$ ) and is not claimed; it does not hold on Qwen or Mistral at the same bar. None of those cells hold on Mistral-7B-Instruct-v0.3. Mistral's with-stem ceiling matches its masked-stem rate, so that miss is a weak solver. Qwen 7B also clears TEKS grade 5 and grade 8 at the registered channel bar on that scorer only; Qwen 14B and Phi-4 do not, so those rows are not claimed. On Qwen 14B the Algebra I leftover is strongest on expression and text options and is weak on numeric options; Phi-4 also clears chance on numeric options (0.366,  $n = 71$ ).

We did not ingest every public paper. English forms from 2019, 2021, and 2022 are the discovery STAAR census. Earlier English STAAR forms scored here are a replication set, not additional census years. Most NYSED 2018 graphic stems, TIMSS 2007, and later restricted TIMSS cycles are out of this file.

Constructed-response items are not headline cells. Scoring guides that reduce to a value are matching; explanations were not graded here.

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## A Implementation details

This appendix records the settings that fix the reported numbers. Nothing here changes a claim in Section 3.

### A.1 Frozen strategy programs

The hash of the frozen rule file, recorded 2026-09-16 20:36:16 UTC before any pass rate was computed, is a0eda4bdb02e54765bb7b49dca5e2e8ea9cd9ebb66de67b2aae074e7d55f0b4.

The distractor-derivation hub builds a directed graph on option values under a fixed error-transform set and picks the unique maximum out-degree, else lower-central among ties. Constraint elimination uses shallow stem keywords: nonnegative counts, probability in  $[0, 1]$ , percent in  $[0, 100]$ , positive magnitudes, angle ranges, and mean or median inside the listed data. Arithmetic closure at depth two is unbound execution of  $+$ ,  $-$ ,  $\times$ ,  $\div$ , squares, and square roots on up to 12 stem numbers, with  $\pi$  and  $ab/2$  on geometry-word stems; it picks the option that first appears, else lower-central among that depth. Backsolving parses a one- or two-variable equation from the stem and picks the unique option that satisfies it. The portfolio runs those four in a fixed order with lower-central among survivors and letter B last.

The machine bar for hub and portfolio is the 95th percentile of 200 random programs of matched complexity, seed 20260916.

The image-channel backsolving addendum was hashed before scoring (sha256 d6ee8caf...). It uses Qwen2-VL-7B-Instruct (Hugging Face revision eed13092...) to transcribe

the displayed equation from a page image; substitution is symbolic, outside the model.

### A.2 Language-model scoring

The scored letter is the option letter with the highest next-token log-probability after the chat template, taking the maximum of the bare letter token and the space-prefixed letter token. The seed is 20260916.

Models and Hugging Face revisions:

- Qwen2.5-7B-Instruct: a09a35458c702b33eeacc393d103063234e8bc28
- Qwen2.5-14B-Instruct: cf98f3b3bbb457ad9e2bb7baf9a0125b6b88caa8
- Mistral-7B-Instruct-v0.3: c170c708c41dac9275d15a8fff4eca08d52bab71
- Phi-4 (microsoft/phi-4): 2db69c1c3e91a05d2c64a3185acfbaf36f744e25

The clean-option filter is mechanical. It keeps 1,496 of 2,245 selected-response items. Of those, 1,487 have a single option-letter key and are scored. The masker withheld nothing on 272 of those 1,487, leaving the primary population of 1,215 items with at least one withheld quantity. The first-run pooled rates on all 1,487 are a reference only.

The masking rules were hashed before scoring. Version 1 is af473bb6cd9d46dc74e034f4460d56a808273fe83ffaf457cd7d122c40ae80e8. Version 2 is 8e6186728bf247d8349f1d94b68280b49eeebac7684c7ebeaf80f29345a43ae5.

### A.3 Searched program class

The feature library holds eleven surface cues, enumerated as conjunctions and signed weights up to complexity 3, plus greedy and evolutionary search; 1,847 programs logged.